Midterm 1

Put your name here

DATA 412/612

2026-10-14

October 14, 2026. You have 75 minutes. Questions 1 to 4 are 80 points. Question 5 is a bonus of 10 points.

Put your name in the subtitle at the top of this file. Write each answer in the chunk under that part. Render to PDF. Submit this `.qmd` and the PDF before the 75 minutes are up.

- You may use your notes and the R help files inside RStudio.
- The only website you may use is https://amousavi1.github.io/612/data-612.html.
- You may not use Google, any other website, or an LLM.
- You may not communicate with anyone else.
- Use the tidyverse where a tidyverse verb exists. Question 1 is a function; base R is appropriate there.
- `library(tidyverse)` covers questions 1 to 4. `mpg` is in ggplot2. `starwars` is in dplyr. `mtcars` is in base R. For the bonus, also load `nycflights13`. `flights` is in that package.
- Either pipe is fine: `|>` or `%>%`.
- Partial credit is given for code that runs. If you are stuck, go to the next part.
- Bonus points replace points missed on questions 1 to 4. The exam score cannot go above 80.

1. A function (20 points)

Write a function of two arguments, `x` and `y`.

- If `x` or `y` is not numeric, or the two lengths differ, stop with the message `x and y must be numeric and the same length`.
- Drop every position where `x` or `y` is not finite. `NA`, `NaN`, `Inf`, and `-Inf` are not finite.
- If fewer than 2 positions remain, return `NA_real_`.
- If the mean of the remaining `y` is 0, return `Inf`.
- Otherwise return the mean of the remaining `x` divided by the mean of the remaining `y`.

Write the function in this chunk, then run it on these four pairs. This chunk already has `#| error: true`, so the last call can stop and the document still renders. Leave that error visible.

- `c(1, NA, 4, Inf)` and `c(2, 3, 6, 1)`
- `c(1, 2)` and `c(0, 0)`
- `c(1, NA_real_)` and `c(NA_real_, 4)`
- `c(1, 2)` and `"a"`

2. Rows, columns, and groups (20 points)

Use `mpg`. Answer each part with one pipeline.

(a) A filtered list (8 points)

Keep cars from `year` 2008 in class `compact` or `midsize` with `hwy` at least 30. How many cars is that? Show the five with the highest `hwy`. Break remaining ties by `cty`, highest first. Keep only `manufacturer`, `model`, `year`, `cyl`, `cty`, and `hwy`.

(b) A grouped summary (7 points)

For each `class`, report the number of cars, the mean of `hwy`, and the median of `cty`. Arrange by mean highway mileage, highest first.

(c) A new column, then the names (5 points)

Add a column `economy` equal to `(cty + hwy) / 2`. Place it immediately after `model`. Drop `trans` and `fl`. Show the column names of the result, in order.

3. Sizes, species, and car names (20 points)

(a) A size column (8 points)

Add a `size` column to `starwars`. Assign each character one of these labels, in this order:

- `"missing"` if `height` is missing
- `"large"` if `height` is at least 200 or `mass` is at least 200
- `"droid"` if `species` is `"Droid"`

- `"other"` otherwise

Report how many characters fall in each size, largest count first.

In one sentence, say what would change about the droid count if `"large"` were not checked before `"droid"`. Name the character.

(b) One summary per species (7 points)

Drop characters whose `species` is missing. Keep species that appear more than once. For each of those species, report the median `height` and the median `mass` (ignore missing values inside the medians), and how many rows that species has. Count every row in the species, including rows whose height or mass is missing. Arrange by that count, largest first.

(c) Color pairs and car names (5 points)

Two short pipelines.

- Drop `starwars` rows where `hair_color` or `skin_color` is missing, then report how many different pairs of those two columns remain.
- `mtcars` stores the car name as a row name. Make that a column called `car`. Keep cars with `mpg` of at least 30. Return `car`, `mpg`, and `cyl`, arranged by `mpg`, highest first.

4. One picture (20 points)

From `mpg`, plot highway mileage against engine displacement.

- Points: map `class` to color inside `geom_point()` only, and set an alpha so overlapping points stay visible.
- One straight smooth in black, dashed, with no gray band. The smooth is not colored by `class`.
- Facet by `drv`.
- Use `theme_bw()`. Label the axes `Displacement (litres)` and `Highway mpg`.

In two sentences, say what happens to highway mileage as displacement increases, and whether that pattern shows up in every `drv` panel. Name one difference you can see between panels.

5. Airline delays by month (10 bonus points)

Use `flights`.

Find the five airlines with the highest average delay. Average `dep_delay` and `arr_delay` together over all of that airline's flights, and ignore missing delays.

For those five airlines only, do that average again for each month. Plot the month on the horizontal axis and the log of the monthly average on the vertical axis. Draw one line for each of those five airlines. Color the lines by airline.

Leave out a month whose average is not positive. The log of a negative number is not defined.